

Rita van der Walt, PhD.

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Summary

Electronic engineer (PhD) specialising in industrial IoT, embedded systems, and data reliability. Architects end-to-end data-driven hardware ecosystems, bridging the gaps among custom PCB/firmware design, cloud data pipelines, and advanced statistical modelling. Managed multi-site sensor fleets over 36 months of continuous operation, processing 2.8M+ data points with strict statistical rigour. Python expert in data processing and sensitivity analysis (open source contributor). Proven ability to translate complex technical diagnostics to strategic stakeholder decision-making, with cross-domain skills in hardware, firmware, and data analytics. Expertise in deriving actionable business insights from raw sensor telemetry using rigorous statistical frameworks (SALib, Python).

Education

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| Stellenbosch University , Electronic Engineering | Jan 2024 – Jan 2026 |
| • Conferred March 2026 | |
| Stellenbosch University , Electronic Engineering | Jan 2022 – Jan 2023 |
| • Upgraded to PhD | |
| Stellenbosch University , Electrical and Electronic Engineering | Jan 2018 – Jan 2021 |
| • Conferred December 2021 | |
| • Capstone project (Distinction, 85%): Designed and deployed a P2P LoRa-based water level monitoring system solution for low-resource farms. | |
| • Delivered a full-stack solution encompassing circuit/RF design, custom PCB assembly, firmware, and a local monitoring dashboard. | |
| • System operated independently of electricity or network coverage. | |

Experience

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| Stellenbosch University , Doctoral researcher (Data systems and policy) | Jan 2024 – Jan 2026
2 years 1 month |
| • Statistical engineering and data analytics: Established a robust data-to-policy framework to quantify infrastructure risk, using Python (SALib) to run Global Sensitivity Analysis (GSA) on a 2.8M+ vector longitudinal dataset and identify key drivers of climate vulnerability. | |
| • Stakeholder management and technical communication: Translated technical diagnostics into actionable guidelines for the City of Cape Town (Environmental Health Dept.) and national media, managing the interface with non-technical decision-makers. | |
| • Infrastructure optimisation and project outputs: Developed a cost-benefit framework for building retrofits to reduce climate-vulnerability of infrastructure. | |
| • This directly justified and resulted in capital upgrades (roof retrofitting) in public sector school buildings. | |
| • Regulatory compliance: Benchmarked indoor environmental quality and sensing device performance against local and international standards (ISO, ASHRAE, WHO, SANS). | |
| Stroom Cargo , Lead telemetry and embedded engineer [Contract basis] | Jan 2026 – present
6 months |
| • E-bike telemetry architecture: Directed the ground-up conceptualisation and stakeholder management for a custom cargo e-bike tracking and battery health management system. | |
| • Full-stack IoT implementation: Designing complete PCB architectures in Altium for GPS logging and motor control, and engineering the device-to-cloud communication pipeline via 1NCE, AWS, and Grafana for mass fleet monitoring and management. | |

Fibertime , Lead embedded hardware engineer [Contract basis] - Scalable security infrastructure: Leading the end-to-end productisation of a home alarm system for mass residential rollout in townships. - Hardware and firmware design: Executing comprehensive hardware development (component selection, PCB design, prototyping, debugging) and writing embedded firmware with MQTT cloud integration to enable remote state management via WhatsApp.	Jan 2026 – present 6 months
Stellenbosch University , Specialised sensing consultant [Project basis, custom IoT solutions] - Mining sector (Saldanha): Designed and deployed custom SD-card-based particulate matter (PM) sensors for air quality compliance monitoring in Saldanha active mining zones. - Environmental research (Forestry): Developed and manufactured custom solar-powered LoRaWAN anemometers to monitor canopy wind dynamics in Coetzenburg Forest. - Residential air monitoring: Designed a LoRaWAN household air pollution sensing network and deployed it in households to quantify health risks from amenity fireplace usage in winter.	Jan 2023 – Jan 2024 1 year 1 month
University of Alicante , Visiting research collaborator (EU Horizon project) - Project REMARKABLE: Rural Environmental Monitoring via Ultra-wide-area Networks. - Coordinated technical alignment on ultra-wide area networks between African and European research teams as the South African representative for Work Packages 3 and 4. - System validation (Resource-constrained environments): Validated frugal IoT sensor solutions designed for harsh operating conditions, ensuring performance met contextual, specific constraint criteria. - Knowledge integration and output: Worked to bridge the contextual gap between African and European sensing applications, resulting in the co-authorship of a conference paper.	May 2023 – July 2023 3 months
Stellenbosch University , Postgraduate researcher (MEng) - Full-stack IoT development: Designed, developed, and deployed a proprietary LoRaWAN sensor node from concept to mass production. - Executed RF matching, custom PCB design, and mechanical enclosure modelling for high-reliability deployment. - Manufacturing lifecycle: Managed the transition from in-house prototyping (SMD soldering, PCB milling) to outsourced volume manufacturing for at-scale deployment. - Network operations: Directed the physical deployment and maintenance of the sensor fleet across multiple sites, optimising gateway uptime and power management for a 24-month data collection period. - Dataset engineering: Engineered data pipelines for the cleaning, validation, and metadata mapping of high-frequency telemetry. - Observational analysis: Provided an evidence base of indoor air environmental exposures in different infrastructure buildings, quantifying exposure levels and severity. - Compliance benchmarking: Conducted baseline environmental auditing, quantifying health, comfort, and cognition risks in educational infrastructure by benchmarking real-world sensor data against building and environmental standards (SANS, WHO, ASHRAE, ISO), forming the basis for subsequent retrofitting strategies.	Jan 2022 – Jan 2023 1 year 1 month
Stellenbosch University , Assistant lecturer (Analogue electronics) Module: Design (E) 344 (Analogue design) - Technical leadership: Managed curriculum delivery and practical assessments for 110+ undergraduate students, focusing on the design of object-avoiding autonomous vehicles. - Mentorship and debugging: Provided support for circuit debugging, designed PCBs for the project, and designed and built test benches for the analogue control systems and circuitry. - Team management: Coordinated a team of student assistants (demis) to manage workload distribution, assessment design, and semester planning.	Jan 2022 – Jan 2022 1 month

South African National Space Agency (SANSA) , Space weather apprentice - International collaboration: Selected for the International Space Weather Camp 2022, with lectures from space agencies from USA, Germany, and South Africa, gaining practical insight into satellite communications and space weather physics. - Completed the practical leg of the space weather camp at the South African National Space Agency (SANSA), with international legs cancelled due to Covid-19 restrictions on international travel. - SDR development: Collaborated with team to build a Software Defined Radio (SDR) capable of capturing live image data from NOAA satellites.	June 2022 – July 2022 2 months
Divigraph (Pty) Ltd. , Graduate hardware intern (Vacation work) 8-week internship - Compliance and verification: Executed hardware unit testing and EMC standards compliance testing, ensuring regulatory adherence for industrial-grade hardware. - Hardware development: Contributed to the iteration of existing hardware designs, focusing on reliability improvements and signal noise reduction.	Dec 2021 – Jan 2022 2 months
Divigraph (Pty) Ltd. , Student intern (Vacation work) 10-week internship - PCB design and component selection: Assisted in hardware design, including schematic capture, component selection, and PCB layout. - Performance recognition: Awarded a full final-year bursary based on technical work delivery during this internship.	Jan 2021 – Mar 2021 2 months

Publications

Time-series distribution-based global sensitivity analysis of weather impacts on classroom indoor temperatures: measurement-driven RC characterisation and retrofit insights. R. van der Walt (SSRN)	Jan 2025
Indoor temperature and CO2 in South African primary school classrooms: Inspecting brick, container, and prefab structures. R. van der Walt (Journal of Cleaner Production)	Jan 2024
Lifting the unseen smokescreen on indoor air quality due to amenity indoor fireplaces in South African homes. R. van der Walt (Energy Research & Social Science)	Jan 2024
Dataset of indoor air parameter measurements on indoor air quality and thermal comfort in South African primary school classrooms of various building infrastructure types. R. van der Walt (Data in Brief)	Jan 2024

Skills

Hardware and embedded engineering

Data science and analytics

IoT infrastructure and cloud services

Development tools and standards

Regulatory compliance and standards

Operations and leadership

Projects

Open-source software: SALib (Python)

Jan 2025 – Jan 2025

Core contribution: Enhanced the DMIM algorithm in SALib, the Python library for sensitivity analysis, refactoring bootstrap implementations to enable processing of real-world, non-uniform empirical data.

Dataset publication: Mendeley Data

Jan 2025 – Jan 2025

Verified telemetry: Published a 24-month longitudinal dataset (T, RH, CO2, PM) across six infrastructure types, with extensive static and time-series metadata, establishing a verified high-frequency benchmark for indoor air environments.

Policy and stakeholder engagement

Jan 2025 – Jan 2025

City of Cape Town, Department of Environmental Health: Invited as expert speaker for World Environmental Health Day 2025. Directly advised municipal stakeholders on indoor air quality standards, influencing local environmental health strategy discussions.

Strategic communication and stakeholder engagement

Jan 2024 – Jan 2024

Technical translation (eNCA, RSG, Cape Talk): Featured subject matter expert on national platforms (live TV, radio, other media). Synthesised complex combustion health risk data into accessible public safety messaging, demonstrating the ability to translate technical nuance for non-technical audiences.

References

Prof. Thinus Booysen

Prof. Sara Grobbelaar